

Fractions (Bhinn)

Aasan Hinglish Notes, Types, Operations aur Practice Questions

Fraction kisi poori cheez ke hisse (part) ko represent karta hai. Is guide me fraction kya hai, uske types, simplest form, comparison aur sabhi operations (jod-ghata-guna-bhaag) simple Hinglish me samjhaye gaye hain, aur end me 22 practice questions (answer key ke saath) diye gaye hain.

1. Fraction kya hota hai?

Definition Fraction kisi cheez ke barabar hisson me se kuch hisson ko batata hai. Ise p/q form me likha jata hai.

Example: $3/4$ ka matlab - poori cheez ko 4 barabar hisson me baanta, aur usme se 3 hisse liye.

Parts:

- Numerator (ansh) = upar wala number (kitne hisse liye). Jaise $3/4$ me 3.
- Denominator (har) = neeche wala number (total kitne hisse). Jaise $3/4$ me 4.
- Denominator kabhi 0 nahi ho sakta.

2. Fractions ke Types

- Proper Fraction: numerator < denominator. Jaise $3/4$, $2/5$.
- Improper Fraction: numerator $\geq$ denominator. Jaise $7/4$, $5/5$.
- Mixed Fraction: whole number + proper fraction. Jaise $1\frac{3}{4}$.
- Like Fractions: same denominator. Jaise $2/5$ aur $3/5$.
- Unlike Fractions: alag denominator. Jaise $1/2$ aur $1/3$.
- Equivalent Fractions: value barabar. Jaise $1/2 = 2/4 = 3/6$.
- Unit Fraction: numerator 1 ho. Jaise $1/5$, $1/8$.

3. Equivalent Fractions aur Simplest Form

Equivalent banane Numerator aur denominator dono ko same number se multiply ya divide karo. Jaise $1/2 = (1 \times 2)/(2 \times 2) = 2/4$.

Simplest Form Numerator aur denominator ko unke HCF se divide karo jab tak aur simplify na ho sake.

- Example: $6/9 \rightarrow$ HCF 3 $\rightarrow (6/3)/(9/3) = 2/3$ (simplest form).

4. Fractions ki Comparison

- Same denominator ho to bada numerator wala fraction bada hota hai. Jaise $3/5 > 2/5$.
- Alag denominator ho to LCM lekar same denominator banao, fir compare karo.
- Same numerator ho to chhota denominator wala bada hota hai. Jaise $1/2 > 1/3$.

5. Fractions par Operations

(a) Addition / Subtraction Same denominator ho to numerators jodo/ghatao, denominator wahi rakho. Alag ho to pehle LCM se same banao.

- Example: $1/2 + 1/3 \rightarrow$ LCM 6 $\rightarrow 3/6 + 2/6 = 5/6$

(b) Multiplication Numerator x numerator, denominator x denominator. Fir simplify karo.

- Example: $2/3 \times 3/4 = 6/12 = 1/2$

(c) Division Doosre fraction ko ulta (reciprocal) karke multiply karo.

- Example: $1/2 \div 1/4 = 1/2 \times 4/1 = 4/2 = 2$

6. Conversions

Improper -> Mixed Numerator ko denominator se divide karo. Quotient = whole part, remainder/denominator = fraction part.

- Example: $7/2 \rightarrow 7 \div 2 = 3$ remainder 1 $\rightarrow 3 \frac{1}{2}$

Mixed -> Improper (whole $\times$ denominator) + numerator, denominator wahi.

- Example: $2 \frac{1}{3} \rightarrow (2 \times 3) + 1 = 7 \rightarrow 7/3$

Fraction -> Decimal Numerator ko denominator se divide karo. Jaise $1/4 = 0.25$.

Quick Revision

Proper: upar < neeche ($3/4$) | Improper: upar $\geq$ neeche ($7/4$)

Add/Subtract: denominator same karo (LCM).

Multiply: seedha upar-upar, neeche-neeche.

Divide: doosre ko ulta karke multiply.

Simplest form: HCF se divide.

Practice Questions (Khud Try Karo!)

In 22 questions ko khud solve karne ki koshish karo. Answers neeche answer key me diye gaye hain.

Q1. $3/4$ me numerator kaunsa number hai?

Q2. $7/4$ kis type ka fraction hai (proper / improper)?

Q3. Kya $2/3$ aur $4/6$ equivalent fractions hain?

Q4. $5/10$ ko simplest form me likho.

Q5. $1/2 + 1/2 = ?$

Q6. $3/5 + 1/5 = ?$

Q7. $1/2 + 1/3 = ?$

Q8. $3/4 - 1/4 = ?$

Q9. $2/3 \times 3/4 = ?$

Q10. $1/2 \div 1/4 = ?$

Q11. $7/2$ ko mixed fraction me likho.

Q12. $2 \frac{1}{3}$ ko improper fraction me likho.

Q13. $1/4$ ko decimal me likho.

Q14. Proper fraction ki definition kya hai?

Q15. Unit fraction kya hota hai? Ek example do.

Q16. Kya $4/8$, $1/2$ aur $3/6$ sab equivalent hain?

Q17. $2/5$ aur $3/5$ me se bada kaunsa hai?

Q18. $1/2$ aur $1/3$ me se bada kaunsa hai?

Q19. $6/9$ ka simplest form kya hai?

Q20. $3/4 \times 4 = ?$

Q21. Mixed fraction kya hota hai? Ek example do.

Q22. 0.5 ko fraction me likho.

Answer Key

- Q1: 3 (upar wala number numerator hota hai)
- Q2: Improper ($7 > 4$)
- Q3: Haan ($2/3 = 4/6$, dono barabar)
- Q4: $1/2$ ($5/10$ ko HCF 5 se divide)
- Q5: 1 ($1/2 + 1/2 = 2/2 = 1$)
- Q6: $4/5$ (same denominator, numerators jode)
- Q7: $5/6$ (LCM 6: $3/6 + 2/6 = 5/6$)
- Q8: $2/4 = 1/2$
- Q9: $6/12 = 1/2$
- Q10: 2 ($1/2 \times 4/1 = 4/2 = 2$)
- Q11: $3 \frac{1}{2}$ ($7 \div 2 = 3$ remainder 1)
- Q12: $7/3$ ($(2 \times 3) + 1 = 7$)
- Q13: 0.25
- Q14: Wo fraction jisme numerator < denominator ho.
- Q15: Numerator 1 wala fraction - jaise $1/5$ ya $1/7$.
- Q16: Haan, sab $1/2$ ke barabar hain.
- Q17: $3/5$ bada hai.
- Q18: $1/2$ bada hai.
- Q19: $2/3$ ($6/9$ ko HCF 3 se divide)
- Q20: 3 ($3/4 \times 4 = 12/4 = 3$)
- Q21: Whole number + proper fraction - jaise $2 \frac{1}{3}$.
- Q22: $1/2$ ($0.5 = 5/10 = 1/2$)